Adaptive Digital Current Mode Controller for DC-DC Converters

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Keywords

Digital control, Converter Control, DC power supply, Adaptive control, Field Programmable Gate Array (FPGA).

Abstract

An adaptive control technique for a digital current mode control of dc-dc converters is proposed in this paper. The adaptive digital controller is designed to maintain the optimum gain-phase of the outer control loop regardless of the changes in the load value. In the control algorithm, the load resistance is calculated using the average current and output voltage values inside the digital processor. Next the calculated value of the load resistance is used to tune the Proportional-Integral (PI) controller such that the requirements for crossover frequency and phase margin values are fulfilled. In order to avoid the calculation burden to slow down the program execution speed, the parameter values of PI controller for each value of the load resistance are inserted into a look-up table inside the digital processor. To verify the proposed technique, the controller is implemented in an FPGA to control a buck converter. The experimental results show the excellent performance of the controller in presence of a load change.

Introduction

In dc-dc converters the Current Mode Control (CMC) provides higher bandwidth and inherent over-current protection compared to voltage mode control [1]. In CMC, the external control loop is a voltage loop with output voltage as feedback and a voltage reference. The inner loop in CMC is current loop in which the inductor current (peak, valley or average value of the inductor) is compared with output of the voltage loop. Implementation of the controllers in discrete-domain has several advantages compared to continuous domain implementation [2]. Discrete-domain (digital) controller performance is less affected by temperature drift or component aging compared to continuous-domain (analog) controller [2]. Digital controllers are also more flexible and easier to update. But they achieve lower bandwidth compared to analog controllers due to delays created by Analog-to-Digital Converter (ADC) and calculations [3]. Different types of solutions are proposed by researchers in order to implement the digital CMC. A predictive digital CMC is proposed in [4]-[6], in these methods the next duty cycle is predicted using the present value of duty cycle, inductor size, one sample of inductor current and input voltage. These methods have dependency on inductor size; the inner current loop in [4]-[5] has inherent delay which results in bandwidth reduction [7]. In order to avoid sampling and measuring the inductor current, the sensor-less CMC have been proposed [8]-[10]. In these techniques, the inductor current is predicted using parasitic parameters. This method requires less hardware components for implementation; on the other hand it has high dependency on temperature and component aging. In another approach, the inductor current is re-generated inside the digital processor

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using the falling slope of inductor current [11]. This solution does not have any dependency on inductor size, but it is mainly applicable for dc-dc converters which are operating at Continuous Conduction Mode (CCM). The common feature of all the proposed methods is dependency of the outer loop transfer function on load resistance value. In these controlled converters, the outer loop transfer function changes when the load value changes. Therefore, the controller should be tuned for worst case condition. As a result, controller cannot always operate in the optimum conditions.

As it was mentioned earlier, digital controller has the merit of flexibility which allows implementation of complicated control algorithms. Therefore, it is possible to implement a compensation mechanism inside the digital controller in order to cancel or minimize the effect of load change in the controlled converter.

In this paper, an adaptive control technique for compensating the load change in digital CMC of the dc-dc converters is proposed. In the proposed solution, the load resistance is calculated using the average value of output current and the output voltage. Normally both of these parameters are available in digital CMC, therefore the proposed solution does not require extra hardware component. The adaptive controller in this solution is a Proportional-Integral (PI) controller with variable parameters. Values of PI controller parameters are determined by load resistance and output filter capacitor.

Digital Peak Current Mode Control

In Fig. 1 the digital Peak Current Mode Control (PCMC) waveforms are shown. The inductor current (I_L) is re-generated inside the digital processor using the inductor current slope measurement technique [11]. The outer voltage loop output is shown with $I_c[n]$. The rising and falling slopes of I_L are shown with $m_r[n]$ and $m_f[n]$ respectively. In PCMC, the switch(es) is turned on in the beginning of the control cycle; it turns off when the inductor current and output of the compensation counter intersect.

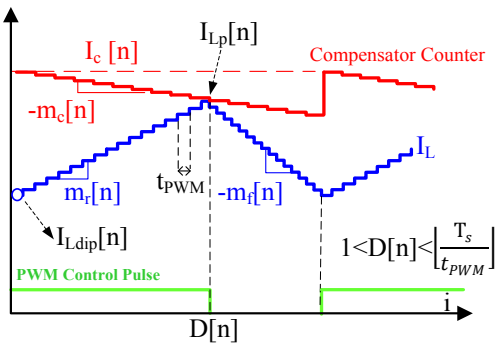

Fig. 1: Digital PCMC waveforms

The compensation counter is replica of compensation ramp which exists in analog PCMC [12]. The compensation slope is shown with $m_c[n]$. The compensation counter (ramp) is required in order to prevent the sub-harmonic oscillation to happen and destabilize the controlled system [12].

The inner loop of digital PCMC block diagram is shown in Fig. 2. In [13], digital PCMC with waveforms shown in Fig. 1 is modeled, which yields

$$\frac{V_o(s)}{I_c(s)} \approx \frac{G_{vd}(s)}{G_{id}(s)R_i\left(\frac{s^2}{{\omega_n}^2}+\frac{s}{Q_n\omega_n}+1\right)} \qquad \omega_n = \sqrt{12}f_s \simeq \pi f_s\,, \qquad Q_n \simeq \frac{2(1+\alpha)}{\pi(1-\alpha)} \tag{1}$$

The $G_{vd}(s)$ and $G_{id}(s)$ are defined as $G_{vd}(s) = \frac{V_o(s)}{d(s)}$ and $G_{id}(s) = \frac{I_L(s)}{d(s)}$. Also α in (1) is equal to $\frac{m_f - m_c}{m_r + m_c}$.

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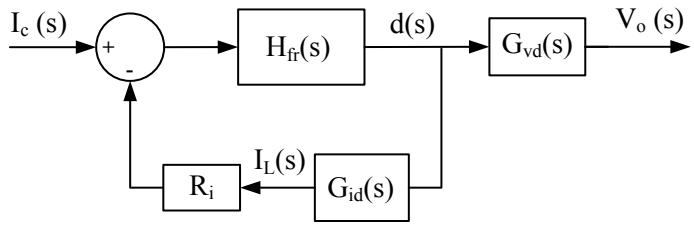
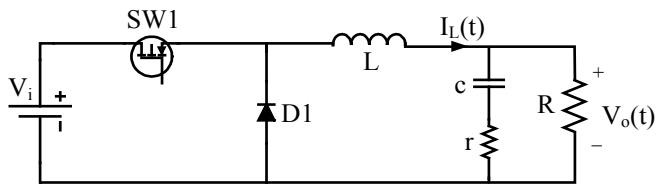


Fig. 2: Inner current loop diagram of digital PCMC

Values of $G_{vd}(s)$ and $G_{id}(s)$ for a buck converter are obtained in [13]. The buck converter and its corresponding transfer functions are shown in Fig. 3.



$$G_{id}(s) = \frac{V_i}{R}\frac{Rcs+1}{Lcs^2 + s\left(\frac{L}{R}+cr\right)+1}$$

$$G_{vd}(s) = V_i\frac{1+rcs}{Lcs^2 + s\left(\frac{L}{R}+cr\right)+1}$$

Fig. 3: Buck converter schematic and its $G_{vd}(s)$ and $G_{id}(s)$

Using transfer function in Fig. 3 and (1) the inner current loop transfer function obtains

$$\frac{V_o(s)}{I_c(s)} \cong \frac{R}{R_i}\frac{(1+rcs)}{(Rcs+1)\left(\frac{s^2}{{\omega_n}^2}+\frac{s}{\omega_n Q_n}+1\right)} \tag{2}$$

In (2) the $\frac{V_o(s)}{I_c(s)}$ has a zero and one first order pole and couple of complex conjugate poles. The first order pole which is $Rcs+1$ is dependent on R and c. The output filter capacitor (c) is a fixed parameter, but the R value which represents the load impedance can change due to the load variation. Normally, the PI controller in the outer loop is tuned such that, the PI zero cancels the pole in inner current loop transfer function [13]. A PI controller has the following transfer function

$$T_{PI}(s) = \frac{I_c(s)}{e(s)} = K_P + \frac{K_I}{s} = \frac{K_P s + K_I}{s} \tag{3}$$

From (3) the $T_{PI}(s)$ has a zero at $-\frac{K_I}{K_P}$. The error signal which is input to the PI controller is shown with e(s). In the common PI controller structures, the K_P and K_I values are constant; they are tuned either for nominal load values or the worst case condition. In analog controllers the PI parameters are represented by capacitor and resistance values, therefore it is difficult to have time variable K_P and K_I. In digital controllers these parameters are coefficients in algebraic controller equation. Using the bi-linear transform $T_{PI}(s)$ can convert to z-domain

$$T_{PI}(z) = \frac{I_c(z)}{e(z)} = \frac{\left(K_P + K_I\frac{T_i}{2}\right) + \left(-K_P + K_P\frac{T_i}{2}\right)z^{-1}}{1-z^{-1}} \tag{4}$$

In (4), T_i is sampling time for conversion of the system from s-domain into z-domain. The $\frac{I_c(z)}{e(z)}$ in (4) can be converted to an algebraic equation

$$I_c[n] = I_c[n-1] + K_1 \cdot e[n] + K_2 \cdot e[n-1] \qquad K_1 = \left(K_P + K_I\frac{T_i}{2}\right), K_2 = \left(-K_P + K_P\frac{T_i}{2}\right) \tag{5}$$

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Adaptive Digital PI Controller

Using (2) the buck converter has a pole at $-\frac{1}{Rc}$ in the inner current loop transfer function. Using the design criteria in which the PI controller zero should cancel the pole in (2), one tuning rule for PI controller obtains

$$\frac{K_I}{K_P} = \frac{1}{Rc} \tag{6}$$

The next tuning rule obtains by keeping the gain constant at crossover frequency (f_c). The open loop transfer function of the outer loop including the PI controllers equal to

$$T_{OP}(s) \cong \frac{R \cdot K_I(sK_P/K_I + 1) \cdot (1 + rcs)}{s \cdot R_i \cdot (Rcs + 1) \cdot \left(\frac{s^2}{{\omega_n}^2} + \frac{s}{\omega_n Q_n} + 1\right)} \tag{7}$$

The load dependent parameters in (7) can be separated as

$$G_R(s) = \frac{R \cdot K_I(sK_P/K_I + 1)}{s \cdot (Rcs + 1)} \tag{8}$$

Substituting (6) in (8) yields

$$G_R(s) = \frac{R \cdot K_I}{s} \tag{9}$$

Assuming $|G_R(2\pi f_c)| = G$, the second tuning rule for PI controller parameters obtains. Using (6) and (9) values of K_P and K_I can be calculated as

$$K_P[n] = R[n] \cdot c \cdot K_I \tag{10}$$

And

$$K_I[n] = \frac{G \cdot 2\pi f_c}{R[n]} \tag{11}$$

Substituting (10) and (11) in (5) yield the required K_1 and K_2 parameters for digital PI controller. The G value can be found by sketching the bode plot of the system using (7) and (8) and finding the required amplitude at f_c.

In Fig.4 the block diagram of adaptive digital PCMC is shown. The "PI parameters calculation" calculates the R using measured I_{Lavg} and V_o values. Additionally, it calculates K_1 and K_2 using (10) and (11). In the buck converter average value of inductor current (I_{Lavg}) and average value of load current in steady-state condition are equal; therefore I_{Lavg} can be used instead of the average load current for finding the load resistance. The measurement gains for output voltage and inductor current are shown by R_V and R_i respectively.

$$R[n] = \frac{R_i V_o[n]}{R_V I_L[n]} \tag{12}$$

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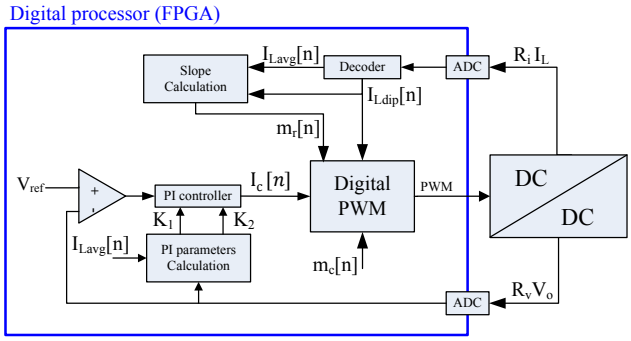

Fig. 4: Block diagram of digital PCMC with adaptive control

The slope calculation block uses the same principles as described in [11] to find the rising inductor current slope ($m_r[n]$). Finding the appropriate value of compensation slope ($m_c[n]$) requires sketching the bode plot of the system using (7). The $m_c[n]$ value does not have significant on crossover frequency, but it changes the phase margin value. Theoretical approach of tuning $m_c[n]$ is explained in [13].

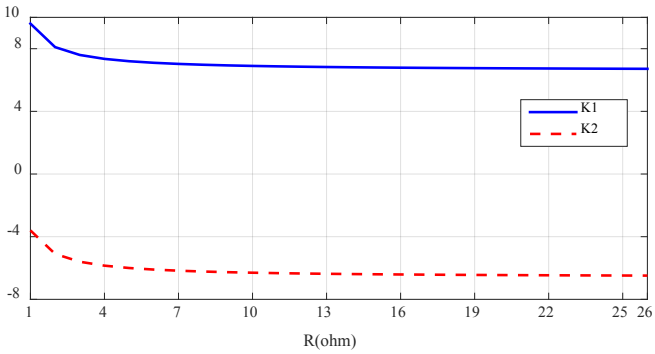

Fig. 5: K_1 and K_2 values versus the load resistance

In order to reduce the calculation time of the digital processor, values of K_1 and K_2 for defined range of R can be calculated offline and saved in a look-up table.

Hardware Implementation and experimental results

The proposed controller is implemented in Xilinx Spartan XI FPGA. A buck converter same as the schematic in Fig. 3 is built to verify the controller strategy. The buck converter specifications are shown in Table I. The switching frequency (f_s) is 50 kHz. The nominal load is 4Ω, the light load resistance is 26Ω. Two 12 bits ADCs are used to sample the inductor current and output voltage. The sampling frequency of these ADCs is equal to switching frequency.

Table I: Buck Converter Parameters

V_i	40 V
V_o	20 V
L	164 µH
C	22 µF
r	10 mΩ
R	4-26Ω
f_s	50 kHz

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In order to evaluate the adaptive controller performance, a constant-parameter PI controller (conventional PI) is also implemented. The tuning criterion for PI controller is defined to have at least 60° of phase margin in worst condition. For conventional type of PI controller, the worst condition is the maximum load resistance (light load). Tuning the conventional PI controller in the light load condition leads to high phase margin in nominal load value. Although the high phase margin value results in a more stable controller, but it will slow down the transient response. Therefore, it is expected that the conventional PI controller has slower transient response than the adaptive PI controller.

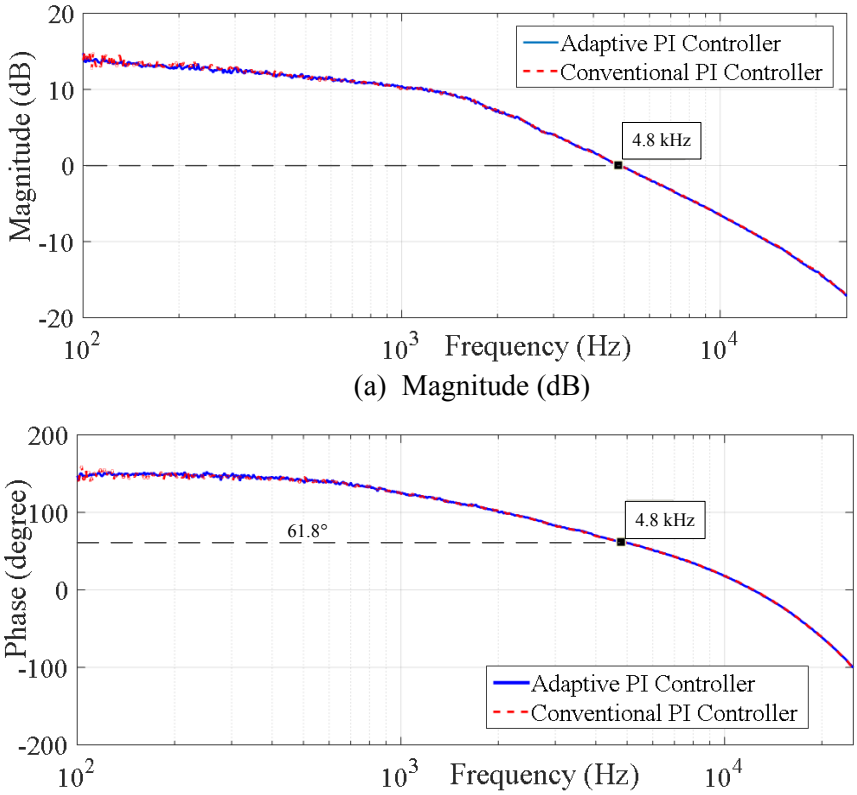
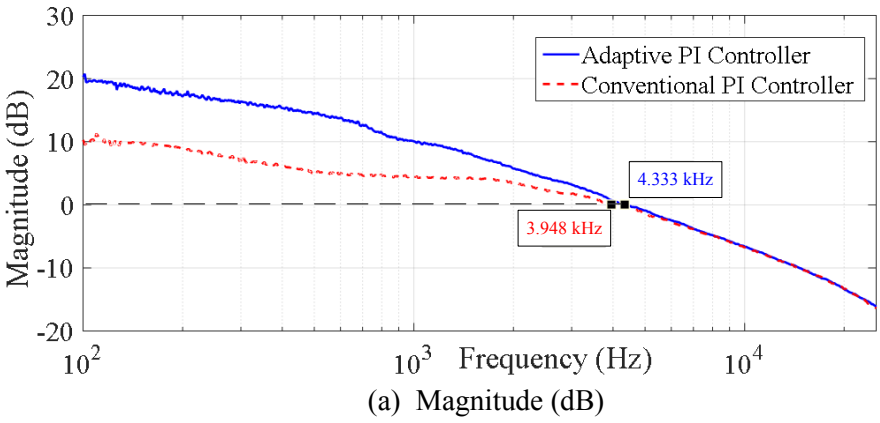


Fig. 6 : Comparison of measured open-loop gain and phase controllers in minimum load of 26Ω

The open-loop gain and phase which is identified by (7) is measured using Agilent 4395A network analyzer and an injection transformer. The measured values are saved in an ASCII file, and then they are plotted in MATLAB. In Fig. 6, the measured open-loop transfer functions using Adaptive and conventional PI controllers in minimum load value are shown. Since the minimum load condition is the worst design case, both of the controllers have equal crossover frequency (≈ 4.8 kHz) and phase margin (≈61.8°).



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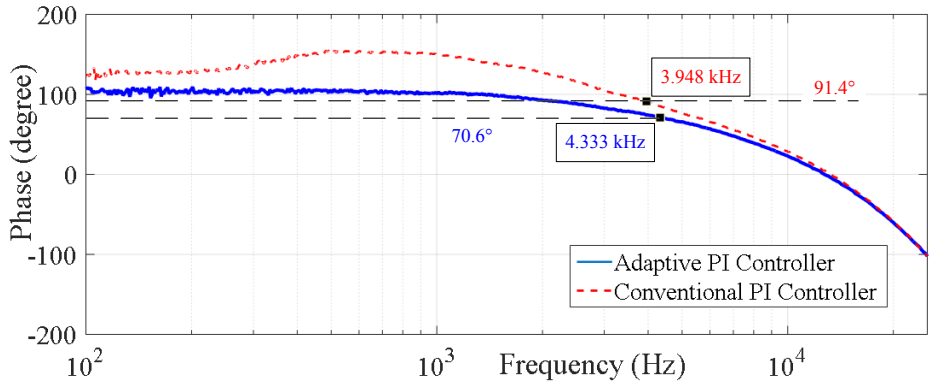
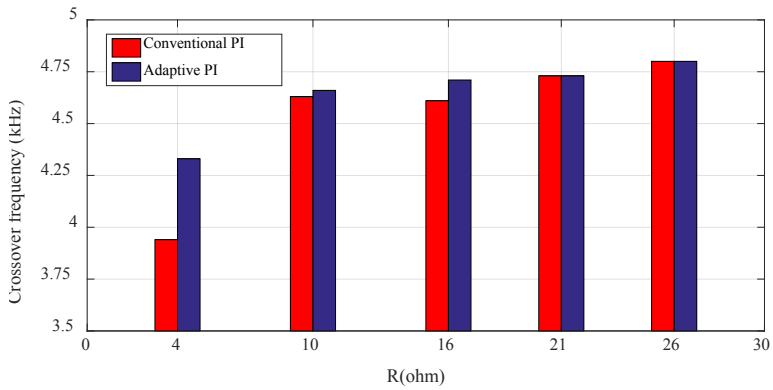


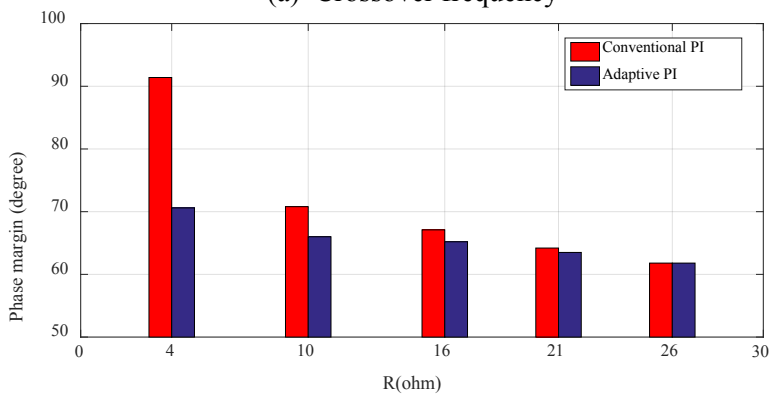
(b) Phase(degree)

Fig. 7: Comparison of measured open-loop gain and phase controllers in nominal load of 4Ω

The $T_{op}(s)$ changes due to the load value change. The adaptive PI controller is able to compensate for the transfer function changes due to the load variation. But in the conventional PI controller, no reaction to the load change occurs. Therefore as Fig. 7 indicates, the crossover frequency declines and phase margin increases in the nominal load condition in the case of conventional PI controller. The phase margin increase and crossover frequency decline result in slower response of conventional controller compared with the adaptive controller. The crossover frequency variation for adaptive controller from the minimum load to nominal load condition is around 467 Hz, whereas this value is 852 Hz for conventional PI controller (almost two times of adaptive controller). In the same manner, the phase margin change for adaptive controller is about 9° versus 30° for conventional PI controller.



(a) Crossover frequency



(b) Phase margin

Fig. 8: Comparison of measured crossover frequency and phase margin in different load values

The measured crossover frequency and phase margin in five load values using conventional and adaptive controllers are shown in Fig. 8. This figure demonstrates that the crossover frequency of conventional PI controller is always lower or at most equal to crossover frequency of adaptive PI

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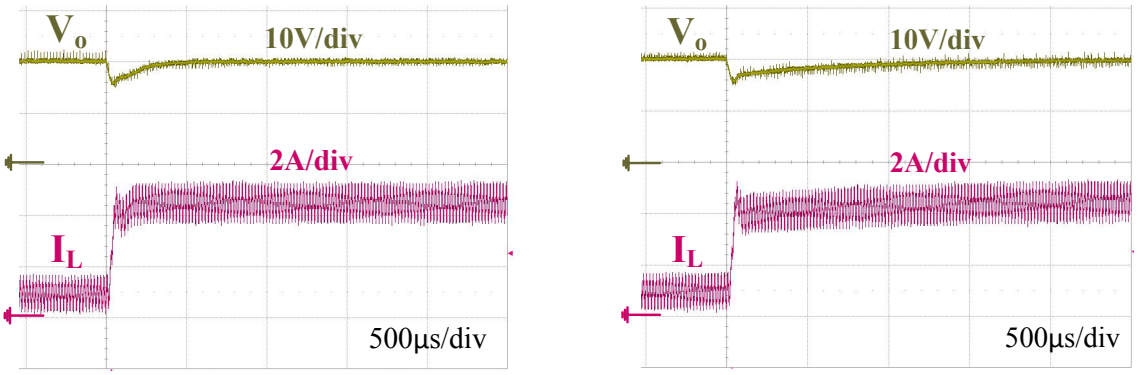
controller. Therefore, adaptive controller has generally faster transient response. Furthermore, phase margin of conventional PI controller is mostly larger or at least equal to adaptive PI controller. The phase margin of the adaptive PI controller is always larger than defined design target of 60°. The higher phase margin value is another reason for slower transient response of the conventional PI controller.

In Fig. 9 transient response of the buck converter using conventional and adaptive PI controllers are compared. The settling time for adaptive controller is around 400 µs versus the 1.6 ms for conventional PI controller. The main reason for the difference is that the conventional PI controller has lower crossover frequency and higher phase margin in nominal load value. When the step change in the load occurs, the load current increases; therefore parameters of adaptive PI controller change in order to compensate the effect of load change.



(a) Adaptive PI controller (b) Conventional PI controller

Fig. 9: Comparison of controller responses for step change in the load from 26Ω to 4Ω

Conclusion

An adaptive digital PI controller for current mode control of dc-dc converters is proposed in this paper. Controller parameters are updated with changes in the load resistance value. The load resistance value is calculated using measured output voltage and inductor current. The adaptive PI controller is designed to maintain the crossover frequency and phase margin almost constant in presence of the load change. The proposed controller is implemented in an FPGA to control a buck converter. The experimental results show the superiority of proposed solution in the term of maintaining crossover frequency and phase margin compared to conventional (constant parameter) PI controller. Additionally, the proposed solution has much faster transient response versus the conventional controller in presence of load step-up disturbance.

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